

Program:	Electrical Engineering	Course Code:	EE223
Duration:	180 Minutes	Semester:	Spring 2018
Paper Date:	29-05-2018	Total Marks:	100
Section:	ALL	Weight:	50%
Exam Type:	Final	Page(s):	5

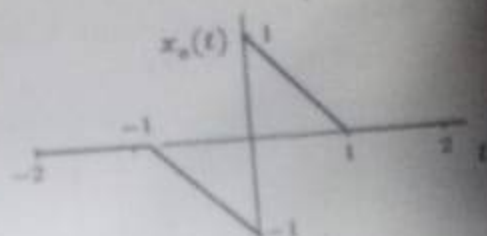
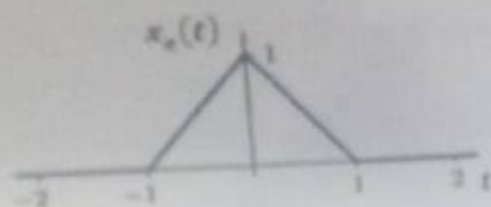
Name: _____ Roll No. _____ Section: _____

Notes: Carefully read the paper and attempt each question (only once). Attempt all parts of a question together.

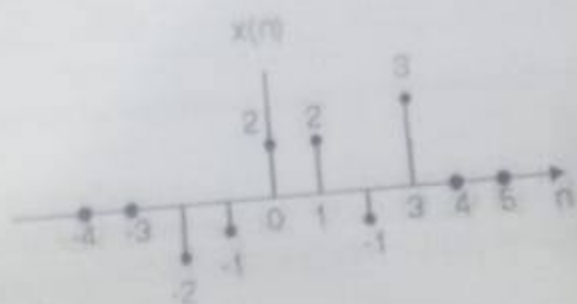
Q: 1 (CLO # 1)

20 Marks

Even and odd components of a continuous time signal $x(t)$ are given below. Sketch the original signal $x(t)$.



b) Determine the power and energy of following signal and state with reason if it is power signal or energy signal.



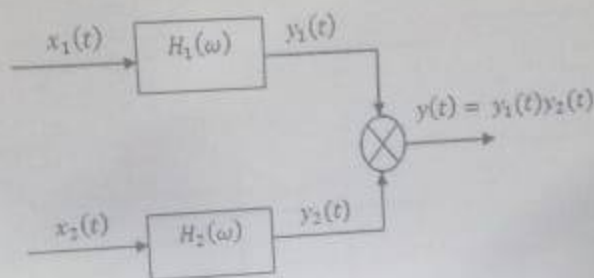
c) Compute the following integral.

$$\int_{-\infty}^{+\infty} e^{j\pi t} \cos\left(\frac{\pi t}{3}\right) \delta(-t + 15) dt$$

Question: 4 (CLO # 4)

15 Marks

Signals $x_1(t) = 10^4 \text{rect}(10^4 t)$ and $x_2(t) = \delta(t)$ are applied at the inputs of the ideal low pass filters $H_1(\omega) = \text{rect}\left(\frac{\omega}{40,000\pi}\right)$ and $H_2(\omega) = \text{rect}\left(\frac{\omega}{20,000\pi}\right)$. The outputs $y_1(t)$ and $y_2(t)$ of these filters are multiplied to obtain the signal $y(t) = y_1(t)y_2(t)$.



- I. Sketch $X_1(\omega)$ and $X_2(\omega)$.
- II. Sketch $H_1(\omega)$ and $H_2(\omega)$.
- III. Sketch $Y_1(\omega)$ and $Y_2(\omega)$.
- IV. Find the bandwidth of $y_1(t)$, $y_2(t)$, and $y(t)$.

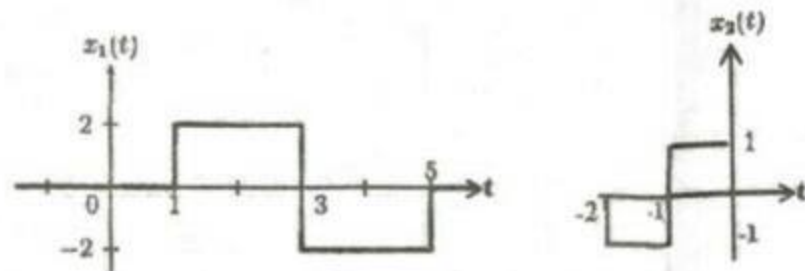
15 Marks

Question: 5 (CLO # 6)

A signal $x(t) = \sin(5\pi t) + \cos(40\pi t)$ is sampled at a rate of 10 Hz.

- a) Find the simplified mathematical expression for the corresponding sampled signal $x(n)$.
- b) Sketch the discrete time signal found in part (a).
- c) Can $x(t)$ be reconstructed by low pass filtering the sampled signal $x(n)$? State your reason clearly.
- d) If we repeat the sampling of above signal $x(t)$ with a frequency of 40 samples per second, is it possible to reconstruct the signal properly? Explain with reason.
- e) What is the minimum sampling frequency required for proper reconstruction of the signal $x(t)$?

Two signals $x_1(t)$ and $x_2(t)$ are shown below:



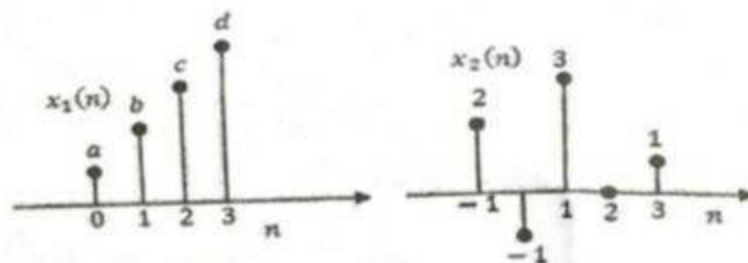
Express $x_2(t)$ as a function of $x_1(t)$ using scale, shift, and / or time-reversal operators.

Question # 2 (CLO 2) ✓

15 Marks

A discrete time signal $x_1(n)$ is convolved with another signal $x_2(n)$ to get the result

$$y(n] = x_1(n) * x_2(n)$$



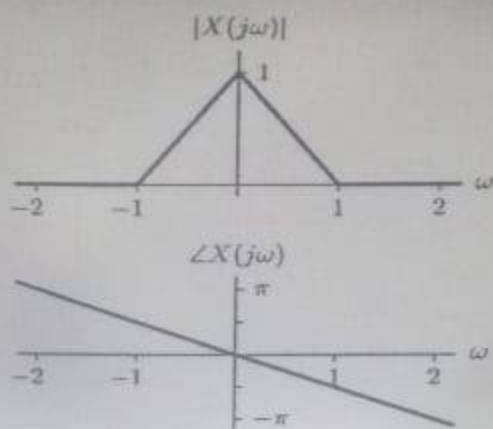
Sketch $y(n)$.

Show all the steps of convolution graphically.

Question: 6 (CLO # 5, PLO#1)

20

The magnitude and phase spectrum of a signal $x(t)$ are given in the following plots



You are required to find the magnitude and phase spectrum of the following five signals i.e. $z_1(t)$, $z_2(t)$, $z_3(t)$, $z_4(t)$ and $z_5(t)$ which are derived from $x(t)$

- a) $z_1(t) = \frac{dx(t)}{dt}$
- b) $z_2(t) = x(t) * x(t)$
- c) $z_3(t) = x(t - \frac{\pi}{2})$
- d) $z_4(t) = x(2t)$
- e) $z_5(t) = x^2(t)$

How to solve:

Six magnitude plots(M1-M6) and six phase plots(A1-A6) are given in Table below

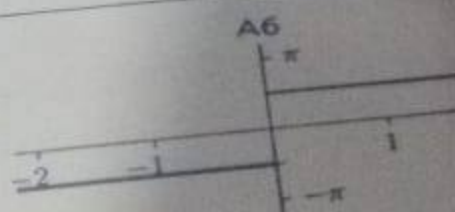
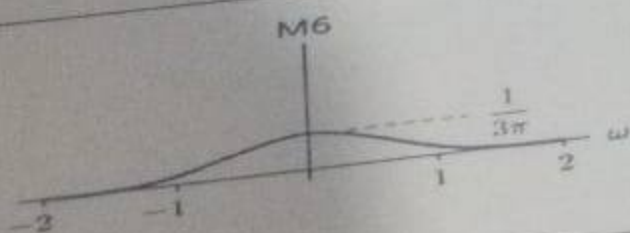
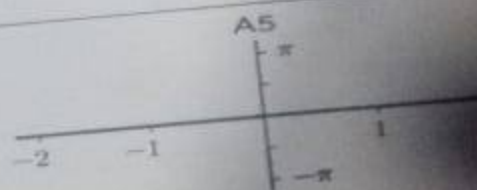
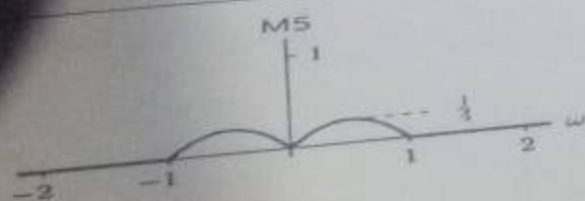
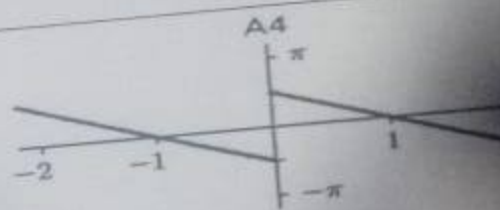
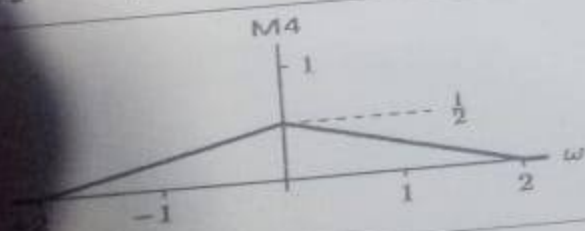
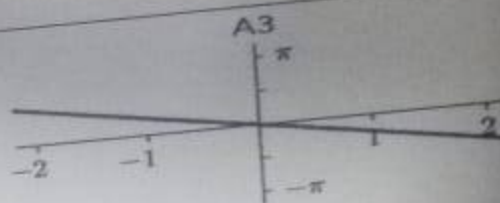
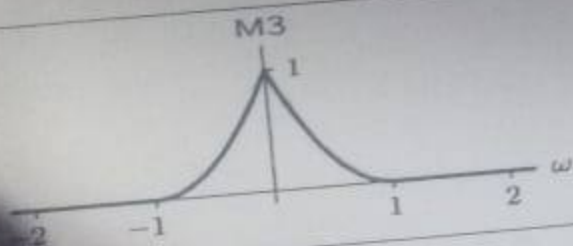
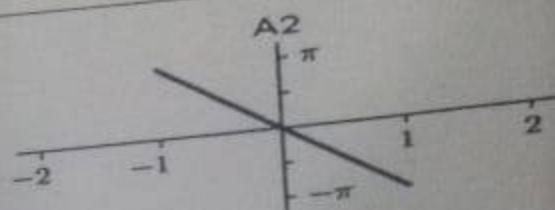
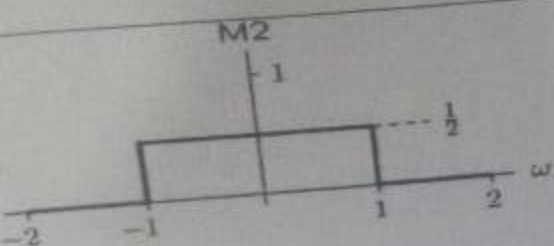
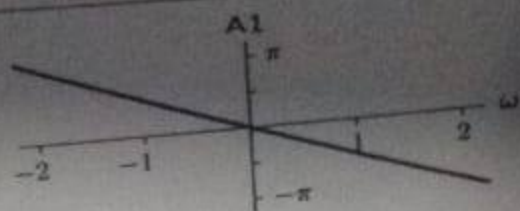
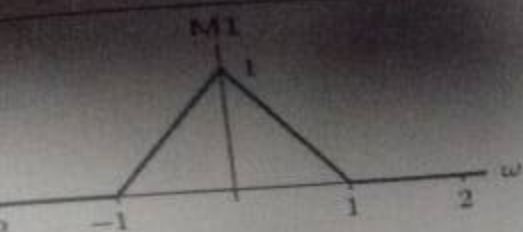
Determine which of these plots is associated with each of the above signals and place the appropriate label (e.g. M7 & A4 for $z_5(t)$).

Also clearly explain the reason for your choice (mathematically as well as in words)

You will get NO CREDIT if you give no explanation.

Note: more than one derived signals can have same magnitude and angle.

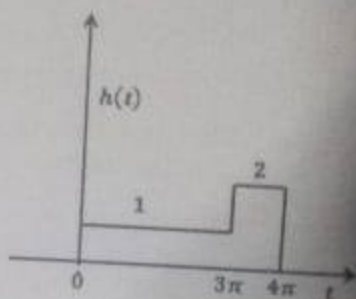
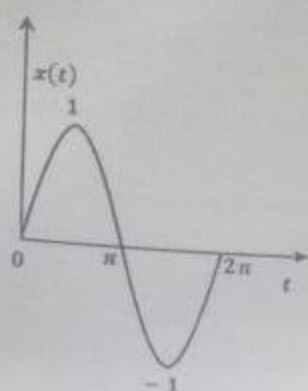
Table 1 for Question 6



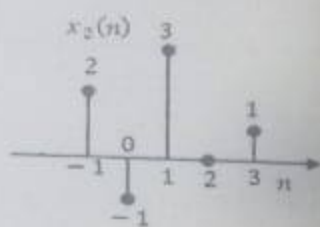
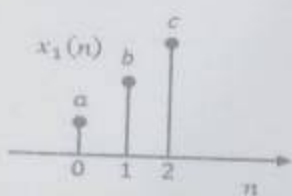
Question: 2 (CLO # 2)

20 M

- a) Two continuous time signals $x(t)$ and $h(t)$ are given below. Using graphical convolution techniques, find the mathematical expression for output such that $y(t) = x(t) * h(t)$.



- b) A discrete time signal $x_1(n)$ is convolved with another signal $x_2(n)$ to get the result $y(n) = x_1(n) * x_2(n)$.



- For what range of 'n', is the output non-zero?
- Find all non-zero sample values of $y(n)$.

Question: 3 (CLO # 7)

10 M

Find the z-transform of the following signal by definition (completely solve) and identify region of convergence. (draw and label ROC graphically)

$$x(n) = \left(\frac{1}{7}\right)^n u(n) - \left(\frac{1}{4}\right)^n u(-n-1)$$

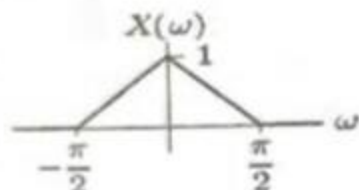
Question # 3 (CLO 3)

The exponential Fourier series of a certain function is given as

$$x(t) = (4 + 3j)e^{-j4t} + j5e^{-j2t} + 2 - j5e^{j2t} + (4 - 3j)e^{j4t}$$

- Sketch the exponential Fourier spectra.
- Sketch the trigonometric Fourier Spectra.
- Find the compact trigonometric Fourier series from the spectra.

Question # 4 (CLO 5)



Sketch and properly label the magnitude and phase spectrum of

- $x(3t)$
- $x(t-5)$
- $x(t) \cos 2\pi t$
- $x(t) * \text{sinc} \frac{\pi t}{4}$

Question # 5 (CLO 6)

Consider sampling of the follow signal at Nyquist rate starting from $t=0$;

$$x(t) = \sin 2\pi t.$$

- Sketch $x(nT)$ and the reconstructed continuous signal $x_r(t)$
- Is $x_r(t) = x(t)$? If yes, state the reason. If no, what should be the minimum sampling frequency to get the proper reconstruction?

Question # 6 (CLO 7)

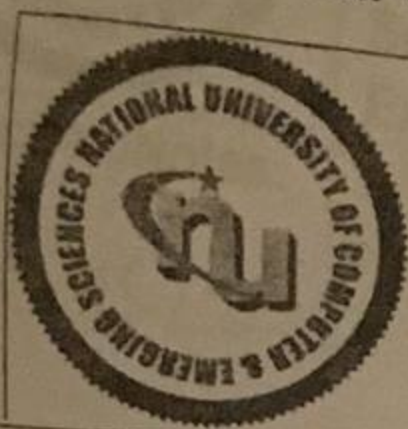
We want to design a **causal discrete time LTI system** with the property that if the input is

$$x(n) = \left(\frac{1}{2}\right)^n u(n) - \frac{1}{4} \left(\frac{1}{2}\right)^{n-1} u(n-1)$$

then the output is $y(n) = \left(\frac{1}{3}\right)^n u(n)$

- Determine the system function $H(z) = \frac{Y(z)}{X(z)}$.
- Find the impulse response $h(n)$ which is inverse z-transform of $H(z)$.

National University of Computer and Emerging Sciences, Lahore Campus



Course: Signals & Systems
Program: BS(Electrical Engineering)
Duration: 60 Minutes
Paper Date: 22nd February, 2017
Section: ALL
Exam: Midterm-I

Course Code: EE206 223
Semester: Spring 2017
Total Marks: 40
Weight: 15%
Page(s): 1
Roll No:

Instruction/Notes: Carefully read the paper and attempt each question (only once). Attempt all parts of a question together.

Question #1: CLO (1)

10 Marks

Mathematical expression of a continuous time signal is given below:

$$x(t) = 2[u(t+6) - u(t+3)] + (-t-1)[u(t+3) - u(t+1)]$$

- Sketch $x(t)$
- Sketch $\frac{1}{2}x(-3t-6)$

Question #2: CLO (1)

10 Marks

Classify the system below as:

- Linear or non-linear
- Time variant or time invariant

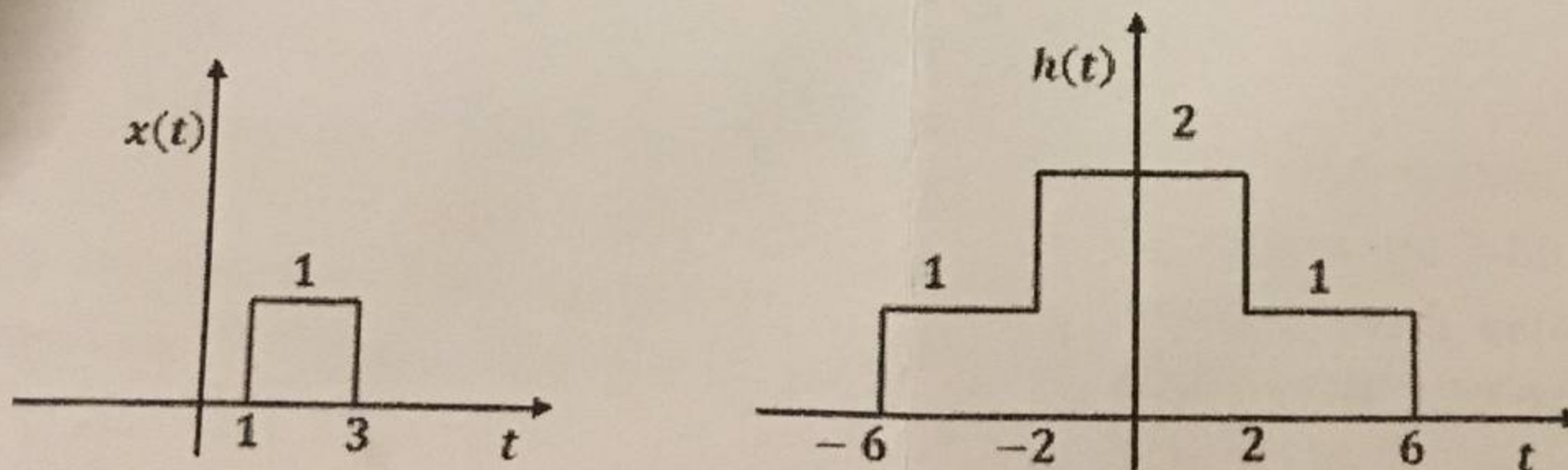
$$y(t) = \cos(x(t))u(t)$$

Provide a concise mathematical justification for your answer.

Question #3 CLO (2)

20 Marks

Two continuous time signals $x(t)$ and $h(t)$ are given below.



Use graphical techniques to find and sketch $y(t) = x(t) * h(t)$.

Handwritten notes and calculations at the bottom of the page include:

- $at - b$
- $t - b$ (circled)
- ab
- $3t + 5$
- $t + 5$ (circled)
- x
- $t + 3$ (circled)
- $t + 5 \cdot (3)$



EE223 Signals and Systems Midterm-2

Time: 60 minutes, Max. Marks: 40, Weightage: 20%, Spring-2015, 05-May

Instructions:

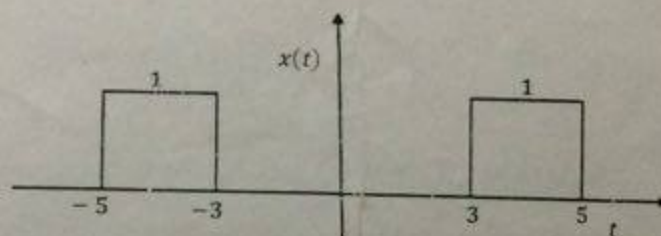
Attempt all the questions.

All questions carry equal marks.

Question # 1:

10 Marks

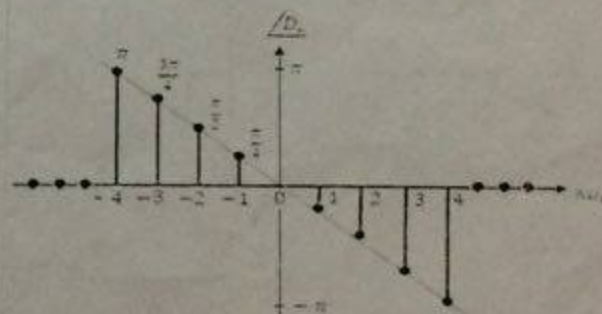
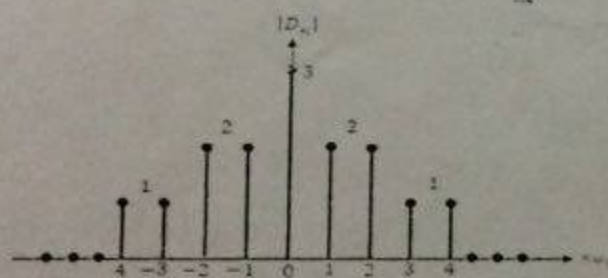
Find the Fourier Transform of following signal.



Question # 2:

10 Marks

Exponential Fourier Spectra of a periodic signal $x(t)$ are given below:

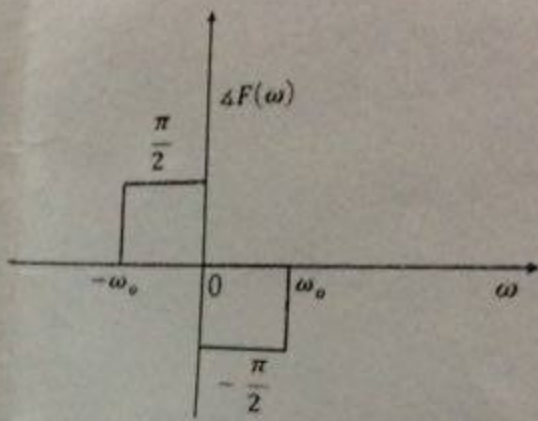
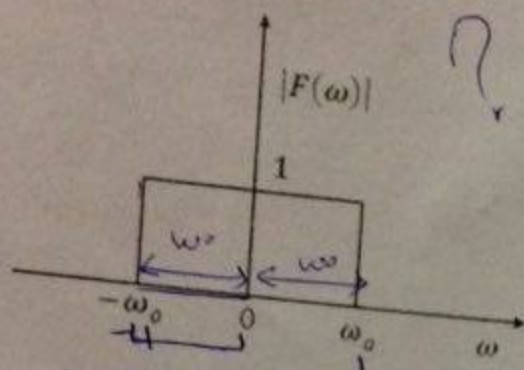


- Find the exponential Fourier series using the above spectra.
- Find the compact trigonometric Fourier series of $x(t)$.
- Sketch the spectra of compact trigonometric Fourier series using part (b).

Question #3 CLO (4)

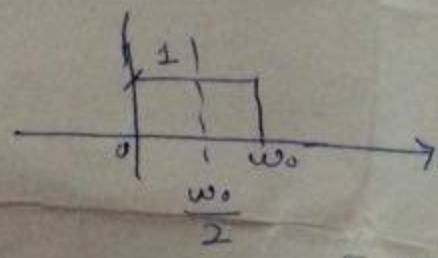
15 Marks

Find and simplify the Inverse Fourier transform $f(t)$ from the magnitude and phase spectra given below



width = $2\omega_0$

ω_0



$$\left[\frac{\omega - \frac{\omega_0}{2}}{2} \right]$$

$$\frac{2\omega - \omega_0}{2}$$

$$\text{rect} = \left[\frac{\omega + \omega_0/2}{2\omega_0} \right]$$

~~2021~~

Question # 3:

Fourier Transform of a continuous time signal $x(t)$ is given below:

$$X(\omega) = 1 - e^{-j\omega}$$

Using properties, find the mathematical expression for the Fourier Transform of following signals.

Note: Clearly explain which property you are using in each part.

- a) $x(t - 6)$
- b) $x(t) \cos 2\pi t$
- c) $x\left(\frac{t}{4}\right)$
- d) $\frac{dx(t)}{dt}$

Question # 4:

10 Marks

A continuous time signal $y(t)$ is defined as $y(t) = x_1(t) * x_2(t)$ where

$$x_1(t) = \text{sinc} 4\pi t \quad \text{and} \quad x_2(t) = \cos 2\pi t$$

- a) Sketch $x_1(t)$ and $X_1(\omega)$
- b) Sketch $x_2(t)$ and $X_2(\omega)$
- c) Sketch $Y(\omega)$

Note: Please label each graph completely.

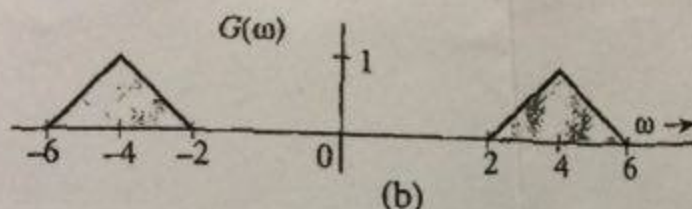


Program: BS(EE)
Duration: 60 Minutes
Paper Date: 10-11-16
Section: ALL
Exam: Midterm-II

Course Code: EE223
Semester: Fall 2016
Total Marks: 50
Weight: 20%
Page(s): 1
Roll No:

Instruction/Notes: Closed book, closed notes, Calculator allowed

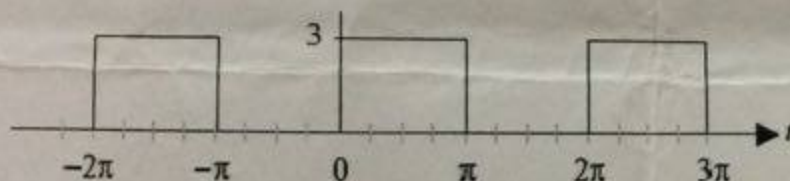
Q.1 (15): $F(\omega) = \delta(\omega - 4) + \delta(\omega + 4)$ and $G(\omega)$ is plotted below;



Plot $H(\omega) = F(\omega) * G(\omega)$

Q.2 (20):

a) Compute the exponential Fourier series of the following function $x(t)$.



b) Find the Fourier coefficients ($a_i, b_i, i=0,1,\dots$) of the following Fourier Series;

$$x(t) = 3 + \cos\left(4t + \frac{\pi}{4}\right) + \sin\left(10t + \frac{\pi}{3}\right).$$

Q.3 (15): The following causal systems are described by input $[x(t)]$ and output $[y(t)]$ equations. Determine if the systems are 1) linear/ Non-linear, 2) time-invariant / time-varying, 3) memory-less / with memory? For each part, write your choices clearly in a box. .

a) $y(t) = x(t - 2)$

b) $y(t) = U(x(t))$

c) $y(t) = tx(t)$

Q.1 (15) $f_1(t) = U(t+1) - U(t-1)$, and $f_2(t) = U(t-1) - U(t-3)$.

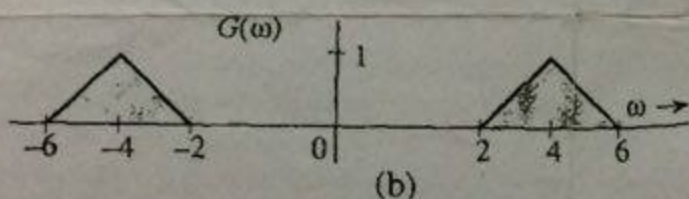
a) Compute $g(t) = f_1(t) * f_2(t)$ by applying time-shift property of convolution and using the fact that $U(t) * U(t) = tU(t)$.

b) Plot $g(t)$ for $-\infty < t < \infty$.

Q.2 (20): Compute the Fourier transform $X(\omega)$ of the following function;

$$x(t) = \begin{cases} \cos \frac{\pi}{2} t, & -1 < t \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

Q.3 (15): Find the inverse Fourier transform $g(t)$ of the following function;



Useful Tables

a) Fourier Transforms

$$\Delta\left(\frac{t}{\tau}\right) \Leftrightarrow \frac{\tau}{2} \text{sinc}^2\left(\frac{\omega\tau}{4}\right)$$

$$\text{rect}\left(\frac{t}{\tau}\right) \Leftrightarrow \tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$$

b) Fourier Transform Properties: If $x(t) \Leftrightarrow X(\omega)$, then

$$X(t) \Leftrightarrow 2\pi x(-\omega)$$

$$x(at) \Leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

$$x(t-t_0) \Leftrightarrow e^{-j\omega t_0} X(\omega)$$

$$x(t)e^{j\omega_0 t} \Leftrightarrow X(\omega - \omega_0)$$

$$x_1(t) * x_2(t) \Leftrightarrow X_1(\omega) X_2(\omega)$$

$$x_1(t)x_2(t) \Leftrightarrow \frac{1}{2\pi} X_1(\omega) * X_2(\omega)$$

$$2t \int x(t) * x(t-\tau) d(t) \int U(t+1) U(t-1) dt$$

$$\approx t U(t-1)$$



Course: Signals & Systems
 Program: BS(Electrical Engineering)
 Duration: 60 Minutes
 Paper Date: 8th April, 2017
 Section: ALL
 Exam: Midterm-II

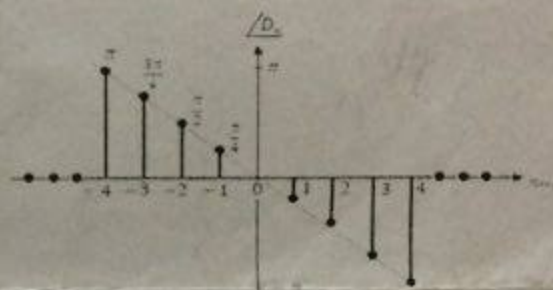
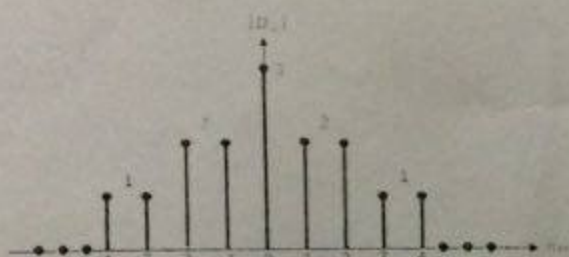
Course Code: EE223
 Semester: Spring 2017
 Total Marks: 40
 Weight: 15%
 Page(s): 1
 Roll No: 15-4522

Instruction/Notes: Carefully read the paper and attempt each question only once. Attempt all parts of a question together.

Question #1: CLO (3)

15 Marks

Exponential Fourier Spectra of a periodic signal $x(t)$ are given below:



- Find the exponential Fourier series using the above spectra.
- Find the compact trigonometric Fourier series of $x(t)$.
- Sketch the spectra of compact trigonometric Fourier series using part (b).
- Find the bandwidth of the signal.

10 Marks

Question #2 CLO (3)

For the following continuous time periodic signals:

- $x(t) = \sin 50\pi t + \cos 30t$
- $y(t) = 3 + 2\cos\left(\frac{5\pi}{2}t\right) + \sin\left(\frac{6\pi}{5}t + \frac{\pi}{3}\right) + 7\cos\left(\frac{\pi}{4}t + \frac{\pi}{4}\right)$ $T=8?$

- Find the fundamental frequency and time period. $1 \text{ } T=2\pi?$
- Identify which harmonics are present in the signal. $\pi/4 \text{ } T=8?$